



上海赫立智能机器有限公司

iHORRY Shanghai iHorry Machine Co.,Ltd.

地址: 上海市松江区玉谷路66号 赫立元吉总部大楼 电话: 021-69186996 传真: 021-69186997 www.ihorry.com

AOI&GRR Report

Machine model: LX535iLS-DM

Serial Number: GRR-LX0959165825-1

Manufacturer: Shanghai iHorry Machine Co., Ltd.

Calibrator: Fang Mingwen

Calibration date: 2026/9/7



GR&R test results	
All detected values for this device are within the acceptable range.	
GR&R for X offset	3.80%
GR&R for Y offset	2.07%
Note: When $GR\&R \leq 10\%$, it indicates that the detection equipment is precise. When $10\% \leq GR\&R \leq 30\%$, it indicates that the detection equipment is correct. When $GR\&R \geq 30\%$, it indicates that the detection equipment is of poor quality.	

Auditor: Fang Mingwen

Approver: Shi Xiaoyu



上海赫立智能机器有限公司

iHORRY Shanghai iHorry Machine Co.,Ltd.

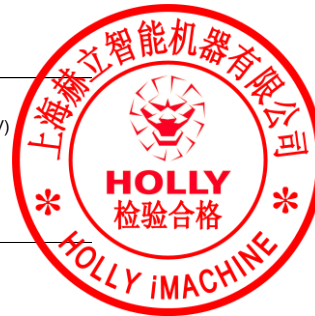
地址: 上海市松江区五谷路66号 赫立元吉总部大楼 电话: 021-69186996 传真: 021-69186997 www.ihorry.com

GR&R - X

Operator	Trial#	Part										AVG
		1	2	3	4	5	6	7	8	9	10	
OP1	1	-0.0300	-0.0140	0.0000	0.0120	0.0010	-0.0020	-0.0180	-0.0130	-0.0430	0.0000	-0.01070
	2	-0.0320	-0.0140	0.0000	0.0120	0.0010	-0.0020	-0.0180	-0.0130	-0.0430	0.0000	-0.01090
	3	-0.0300	-0.0140	0.0000	0.0120	0.0020	-0.0020	-0.0180	-0.0100	-0.0430	0.0000	-0.01030
	Averag	-0.0307	-0.0140	0.0000	0.0120	0.0013	-0.0020	-0.0180	-0.0120	-0.0430	0.0000	XA -0.01063
	Range	0.00200	0.00000	0.00000	0.00000	0.00100	0.00000	0.00000	0.00300	0.00000	0.00000	RA 0.00060
OP2	1	-0.0300	-0.0130	0.0000	0.0120	0.0000	-0.0010	-0.0170	-0.0160	-0.0430	0.0000	-0.01080
	2	-0.0300	-0.0150	0.0000	0.0120	0.0000	-0.0010	-0.0250	-0.0100	-0.0430	0.0000	-0.01120
	3	-0.0300	-0.0130	0.0000	0.0120	0.0000	-0.0010	-0.0180	-0.0100	-0.0430	0.0000	-0.01030
	Averag	-0.0300	-0.0137	0.0000	0.0120	0.0000	-0.0010	-0.0200	-0.0120	-0.0430	0.0000	XB -0.01077
	Range	0.00000	0.00200	0.00000	0.00000	0.00000	0.00000	0.00800	0.00600	0.00000	0.00000	RB 0.00160
OP3	1	-0.0300	-0.0120	0.0000	0.0120	0.0000	-0.0020	-0.0220	-0.0160	-0.0430	0.0000	-0.01130
	2	-0.0300	-0.0140	0.0000	0.0120	0.0010	-0.0020	-0.0180	-0.0130	-0.0430	0.0000	-0.01070
	3	-0.0300	-0.0140	0.0000	0.0120	0.0010	-0.0020	-0.0180	-0.0130	-0.0430	-0.0010	-0.01080
	Averag	-0.0300	-0.0133	0.0000	0.0120	0.0007	-0.0020	-0.0193	-0.0140	-0.0430	-0.0003	XC -0.01093
	Range	0.00000	0.00200	0.00000	0.00000	0.00100	0.00000	0.00400	0.00300	0.00000	0.00100	RC 0.00110
Part AVG		-0.03022	-0.01367	0.00000	0.01200	0.00067	-0.00167	-0.01911	-0.01267	-0.04300	-0.00011	X -0.01078

Rp=	Max(Part AVG) - Min(Part AVG)	=	0.05500
R=	(RA + RB + RC) / No of Operators	=	0.00110
XDIFF=	Max(X) - Min(X)	=	0.00030
UCL=	R * D4	=	0.00284
LCL=	R * D3	=	0.00000
If Trials=3, D4=2.58, D3=0;			
If Trials=2, D4=3.27, D3=0. UCL is R's limit,			

Measurement Unit Analysis				% Total Variation (TV)	
Repeatability-Equipment Variation (EV)					
EV	= R * K1	Trials K1		EV%	= 100 (EV/TV)
		2 0.8862			
		3 0.5908			
Reproducibility-Appraiser Variation (AV)					
AV	= sqrt[(XDIFF * K2)^2 - (EV^2 / (n*r))]			AV%	= 100 (AV/TV)
		= 0.00010			
n = Parts r = Trials		Operators	2 3		
		K2	0.7071 0.5231		
Repeatability & Reproducibility					
GRR	= sqrt(EV^2 + AV^2)	Parts K3		GRR%	= 100 (GRR/TV)
		2 0.7071			
		3 0.5231			
Part Variation (PV)					
PV	= Rp * K3	4	0.4467	PV%	= 100 (PV/TV)
		5	0.4030		
		6	0.3742		
		7	0.3534		
Total Variation (TV)					
TV	= sqrt(GRR^2 + PV^2)	8	0.3375	ndc	= 1.41 (PV/GRR)
		9	0.3249		
		10	0.3146		



赫立经营理念: 勇于创新、锐意进取、诚信合作, 赫立愿与您携手共进、和谐共赢。



上海赫立智能机器有限公司

iHORRY Shanghai iHorry Machine Co.,Ltd.

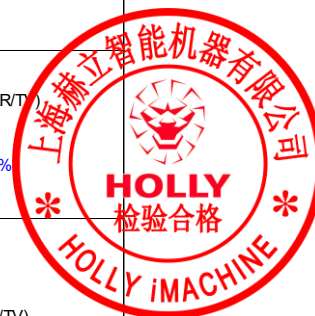
地址: 上海市松江区玉谷路66号 赫立元吉总部大楼 电话: 021-69186996 传真: 021-69186997 www.ihorry.com

GR&R - Y

Operator	Trial#	Part										AVG
OP1	1	0.0230	-0.0100	0.0000	0.0070	0.0290	0.0120	0.0200	0.0220	0.0000	0.0030	0.01060
	2	0.0230	-0.0100	0.0000	0.0070	0.0300	0.0110	0.0210	0.0210	0.0000	0.0030	0.01060
	3	0.0230	-0.0100	0.0000	0.0070	0.0300	0.0120	0.0210	0.0220	0.0000	0.0030	0.01080
	Averag	0.0230	-0.0100	0.0000	0.0070	0.0297	0.0117	0.0207	0.0217	0.0000	0.0030	XA 0.01067
	Range	0.00000	0.00000	0.00000	0.00000	0.00100	0.00100	0.00100	0.00100	0.00000	0.00000	RA 0.00040
OP2	1	0.0230	-0.0120	0.0000	0.0070	0.0290	0.0120	0.0200	0.0220	0.0000	0.0030	0.01040
	2	0.0230	-0.0120	0.0000	0.0070	0.0300	0.0120	0.0200	0.0220	0.0000	0.0030	0.01050
	3	0.0230	-0.0100	0.0000	0.0070	0.0300	0.0120	0.0200	0.0220	0.0000	0.0030	0.01070
	Averag	0.0230	-0.0113	0.0000	0.0070	0.0297	0.0120	0.0200	0.0220	0.0000	0.0030	XB 0.01053
	Range	0.00000	0.00200	0.00000	0.00000	0.00100	0.00000	0.00000	0.00000	0.00000	0.00000	RB 0.00030
OP3	1	0.0230	-0.0120	0.0000	0.0070	0.0290	0.0120	0.0200	0.0220	0.0000	0.0030	0.01040
	2	0.0230	-0.0100	-0.0010	0.0070	0.0290	0.0130	0.0200	0.0220	0.0000	0.0030	0.01060
	3	0.0240	-0.0100	0.0000	0.0070	0.0290	0.0120	0.0200	0.0230	0.0000	0.0030	0.01080
	Averag	0.0233	-0.0107	-0.0003	0.0070	0.0290	0.0123	0.0200	0.0223	0.0000	0.0030	XC 0.01060
	Range	0.00100	0.00200	0.00100	0.00000	0.00000	0.00100	0.00000	0.00100	0.00000	0.00000	RC 0.00060
Part AVG		0.02311	-0.01067	-0.00011	0.00700	0.02944	0.01200	0.02022	0.02200	0.00000	0.00300	X 0.01060

Rp=	Max(Part AVG) - Min(Part AVG)	=	0.04011
R=	(RA + RB + RC) / No of Operators	=	0.00043
XDIF=	Max(X) - Min(X)	=	0.00013
UCL=	R * D4	=	0.00112
LCL=	R * D3	=	0.00000
If Trials=3, D4=2.58, D3=0;			
If Trials=2, D4=3.27, D3=0, UCL is R's limit,			

Measurement Unit Analysis				% Total Variation (TV)	
Repeatability-Equipment Variation (EV)					
EV	=	R * K1	Trials	K1	EV% = 100 (EV/TV)
	=	0.00026	2	0.8862	= 2.02837%
			3	0.5908	
Reproducibility-Appraiser Variation					
AV	=	sqrt[(XDIFF * K2)^2 - (EV^2 / (n*r))]			AV% = 100 (AV/TV)
	=	0.00005			= 0.41015%
n = Parts r = Trials		Operators	2	3	
		K2	0.7071	0.5231	
Reproducibility-Appraiser Variation (GRR)					
GRR	=	sqrt(EV^2 + AV^2)			GRR% = 100 (GRR/TV)
	=	0.00026			= 2.06942%
Part Variation (PV)					
PV	=	Rp * K3			PV% = 100 (PV/TV)
	=	0.01262			= 99.97859%
Total Variation (TV)					
TV	=	sqrt(GRR^2 + PV^2)			ndc = 1.41 (PV/GRR)
	=	0.01262			= 68



赫立经营理念: 勇于创新、锐意进取、诚信合作, 赫立愿与您携手共进、和谐共赢。